Introduction to Model Deployment

Learning Objectives

By the end of this session, you should be able to:

- Understand the need for model deployment and its role in operationalizing Machine Learning solutions.
- Implement serialization techniques to efficiently store and transfer trained models.
- Develop API endpoints to facilitate seamless interaction with deployed models.
- Manage dependencies and hosting environments to ensure reliable and scalable model deployment.

Agenda

In this session, we'll discuss:

- The Need for Model Deployment
- Introduction to Model Deployment
- Serialization
- Introduction to APIs
- Endpoints
- Handling Dependencies
- Hosting a Deployed Model
- Architecture of Model Deployment

Video Break

Common Problems Solved with Machine Learning

Customer Churn Prediction

Predicting which customers are likely to churn based on usage patterns and complaints

Common Problems Solved with Machine Learning

Customer Churn Prediction	Predicting which customers are likely to churn based on usage patterns and complaints
Predictive Maintenance	Identifying machinery components likely to fail before they actually do based on sensor data

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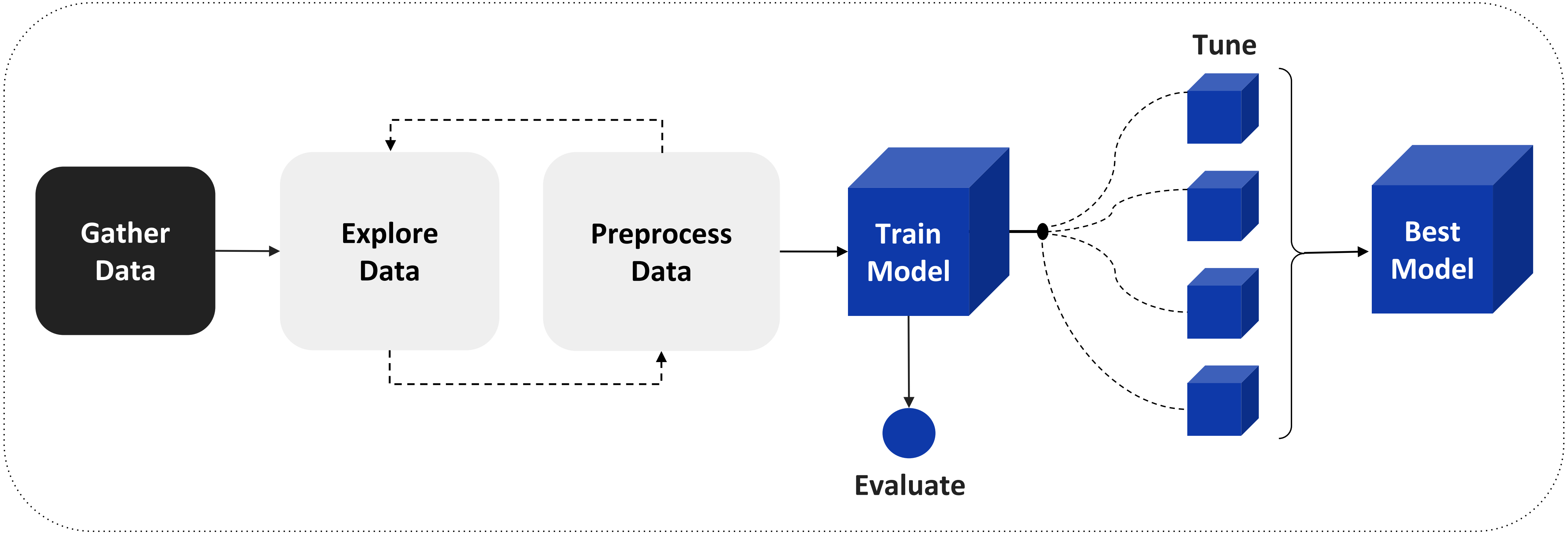
Customer Churn Prediction	**Predicting which customers are likely to churn based on usage patterns and complaints**
Predictive Maintenance	**Identifying machinery components likely to fail before they actually do based on sensor data**
Fraud Detection	**Detecting fraudulent transactions based on historical transaction data**

Common Problems Solved with Machine Learning

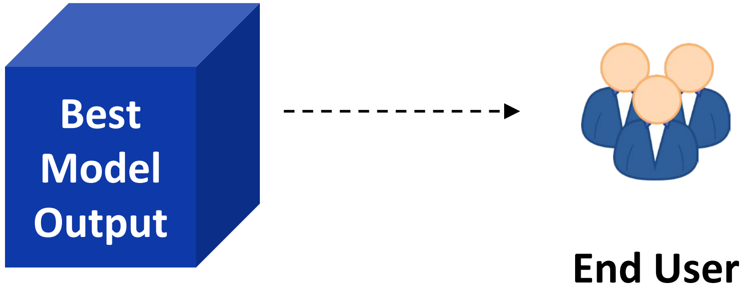
Customer Churn Prediction	Predicting which customers are likely to churn based on usage patterns and complaints
Predictive Maintenance	Identifying machinery components likely to fail before they actually do based on sensor data
Fraud Detection	Detecting fraudulent transactions based on historical transaction data
Personalised Recommendations	Suggesting products or content based on user demography and consumption patterns

Machine Learning Model Building Process



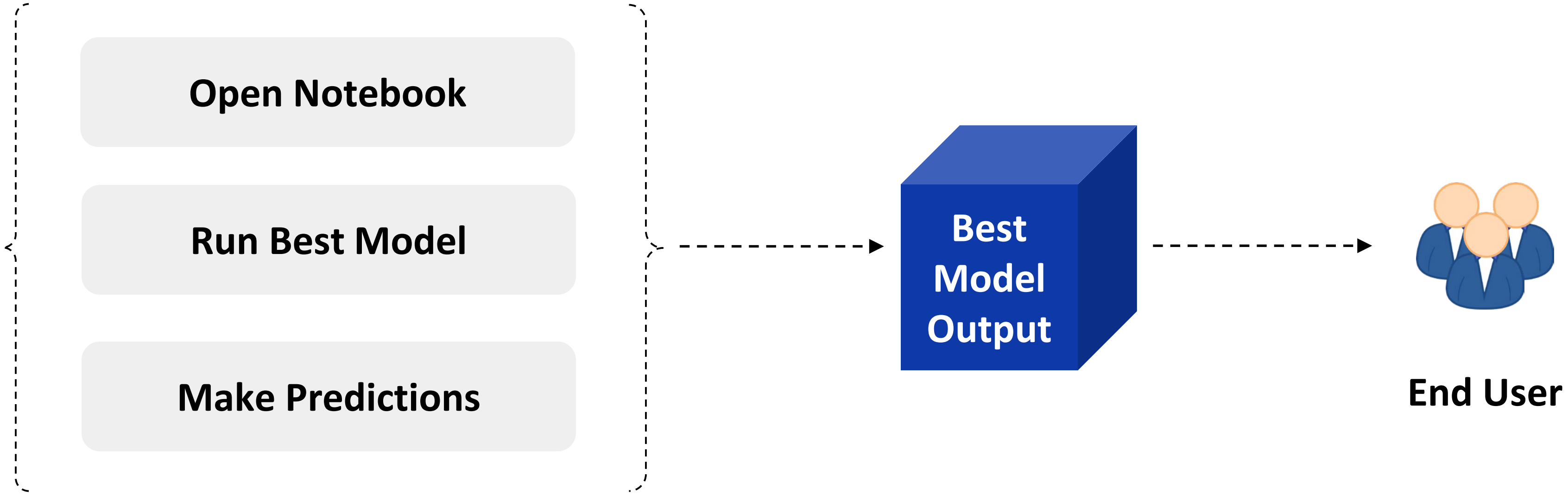
Model Output Consumption

- Need to **expose the best model's output to the end user** for consumption.
- How do we do this?



Model Output Consumption

- Need to **expose the best model's output to the end user** for consumption.
- A **simple approach** can be to **share the Python notebook** with the end user.



Challenges with Sharing Python Notebooks

- Consider the example of a content recommendation model for an online streaming service trained on user watch history.

Accessibility

- How will end users **access the best model** from the data scientist's Python notebook?
- The Python notebook is a set of codes and the end user might not necessarily have the relevant expertise

Users need a **simpler method** to **receive recommendations** as per their preferences without being exposed to associated technicalities.

Challenges with Sharing Python Notebooks

- Consider the example of a content recommendation model for an online streaming service trained on user watch history.

Accessibility

Configuration

- The best model relies on specific dependencies (libraries, packages, and more).
- How do we ensure that the **same "setup" is available on the user's end**?

Even if a user streams from different devices, we **need to ensure dependencies are taken care for smoothly functionality**.

Challenges with Sharing Python Notebooks

- Consider the example of a content recommendation model for an online streaming service trained on user watch history.

Accessibility

Configuration

Performance

- The best model must analyze large amounts of data to identify patterns and generate relevant output.
- Will the Python notebook on the data scientist's system be able to **generate recommendations fast enough**?

If the **recommendations aren't shown at the right time**, then **user engagement will reduce** and consequently impact revenue.

Challenges with Sharing Python Notebooks

- Consider the example of a content recommendation model for an online streaming service trained on user watch history.

Accessibility

Configuration

Performance

Scalability

- The best model is in a single notebook on the data scientist's local system.
- How will it **handle the requirements of multiple consumers**, each expecting instant recommendations?

If the model **cannot provide recommendations to multiple users**, it can lead to a **suboptimal user experience** for those left out, resulting in lower retention rates.

Video Break

Addressing Notebook Sharing Challenges

- To make the model accessible and usable, we need:

1	2	3	4
A way to **share** the model	A way to **ensure consistent configurations**	**Interface for communication** to get fast predictions	**Concurrent access** for multiple users

Addressing Notebook Sharing Challenges

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Sharing and Consistency

“Package” the model and dependencies

Speed and Scalability

“Set” it up for **fast** and **concurrent “usage”**

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That's where **model deployment** comes in.

Model Deployment

Deploying a model means taking a trained ML model, **packaging** it, and setting it up for **inference**.

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1. How do we "**package**"?
2. How do we "**set up**" for **inference**?

Video Break

Addressing Notebook Sharing Challenges

A way to **share** the model

1

A way to **ensure consistent configurations**

2

Interface for communication to get fast predictions

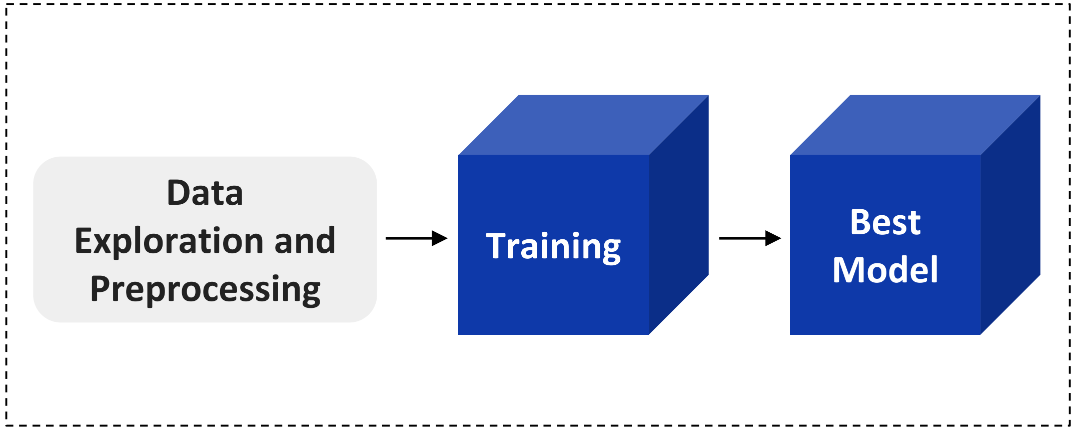
3

Concurrent access for multiple users

4

- Let's start with how we can "**share**" **the best model**.

Current Setup



Local System

- Recall that all the steps involved in the model building process were being executed in a Python notebook on the data scientist's local system.

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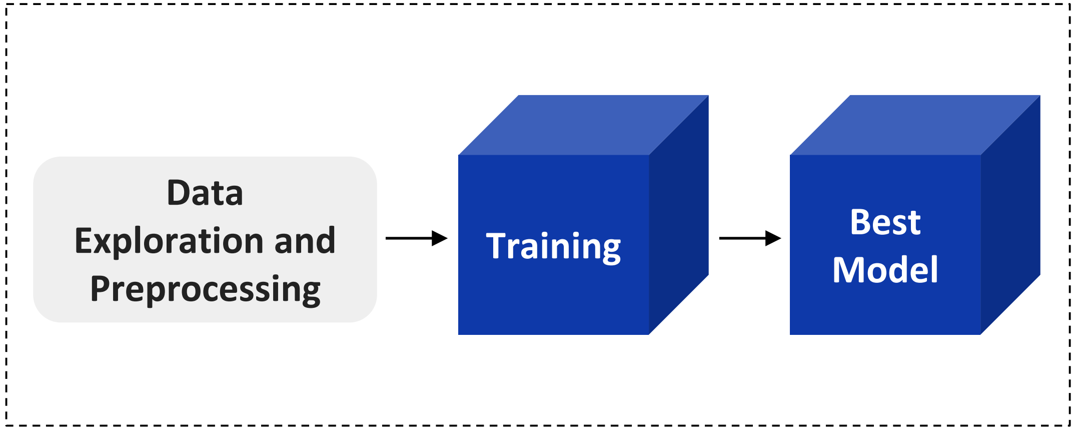

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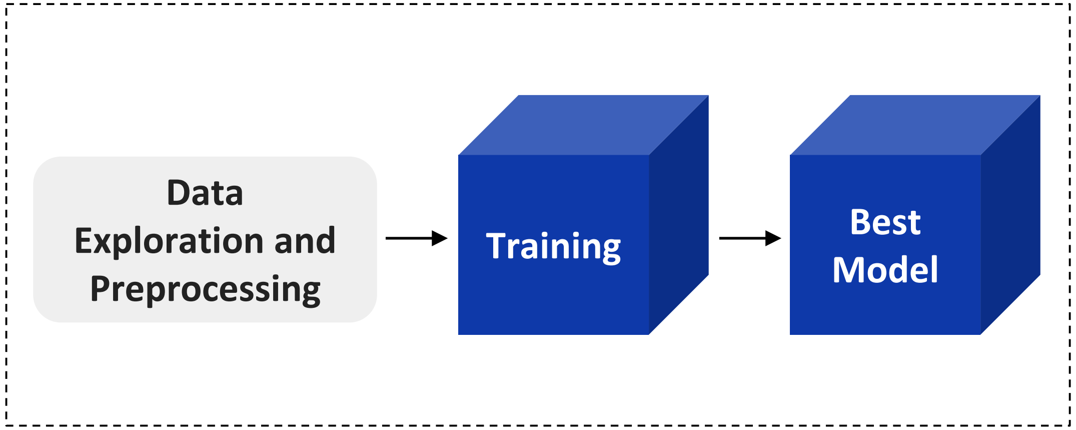

Local System

- Recall that all the steps involved in the model building process were being executed in a Python notebook on the data scientist's local system.
- The notebook can be created in any IDE (Integrated Development Environment).
- Depending on the IDE used, there'll be a specific configuration required to execute the relevant code.

This configuration under which the model was built is an **environment**.

Environment

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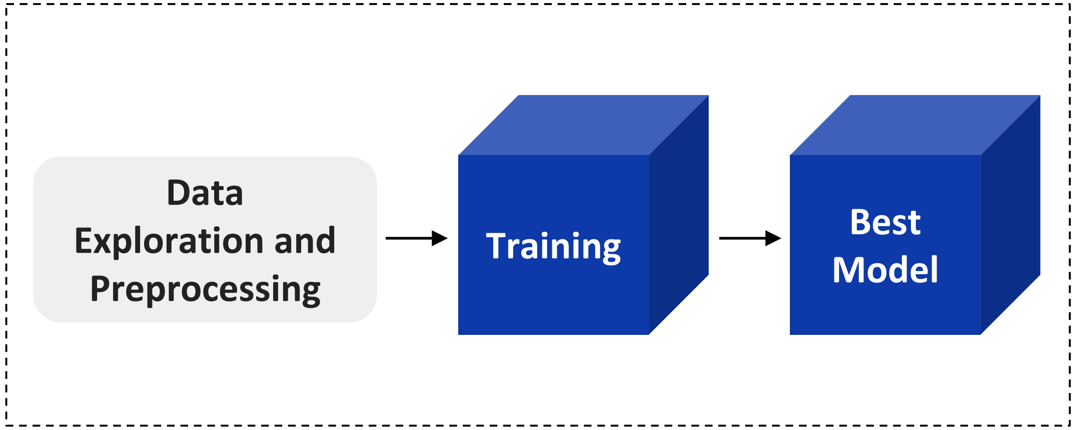
Development Environment

Primarily used for training ML models by hyperparameter tuning.

Production Environment

Primarily used to deploy the final ML model to serve end users.

The Need for Serialization



Development Environment

- The best model is in the development environment and we need to make predictions for new data.

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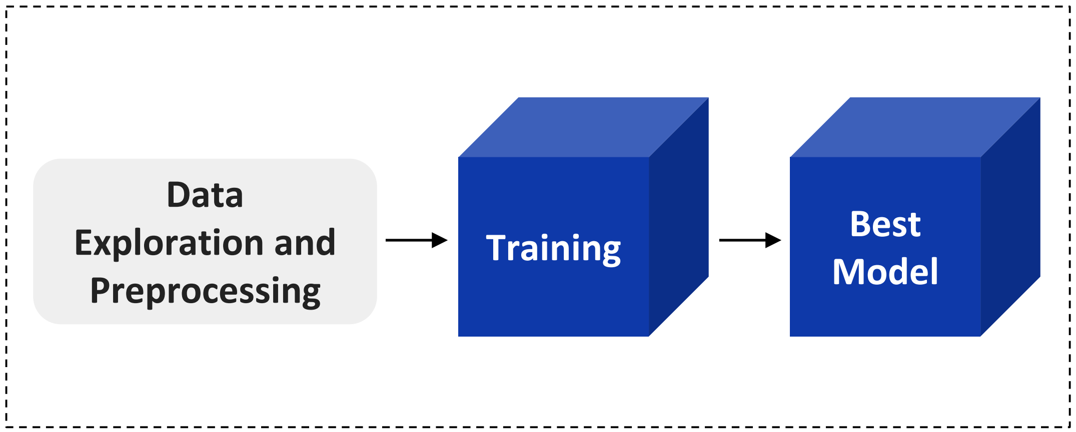

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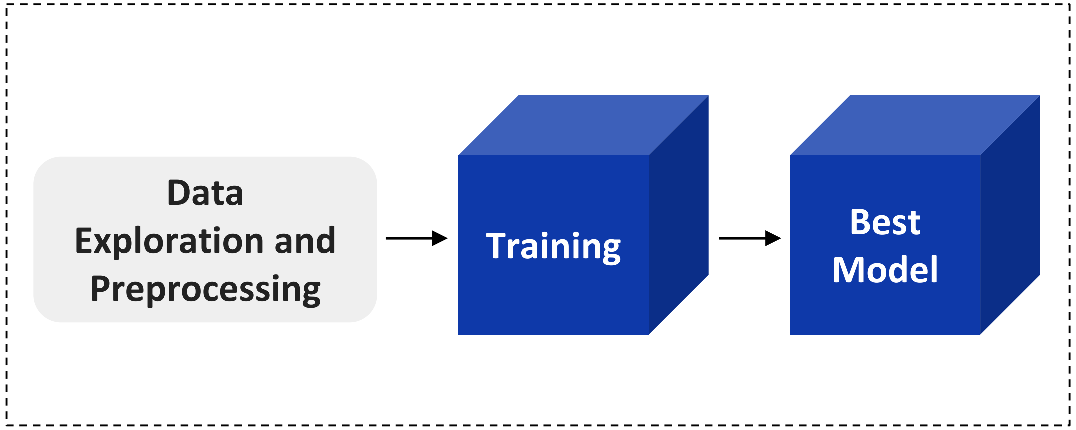

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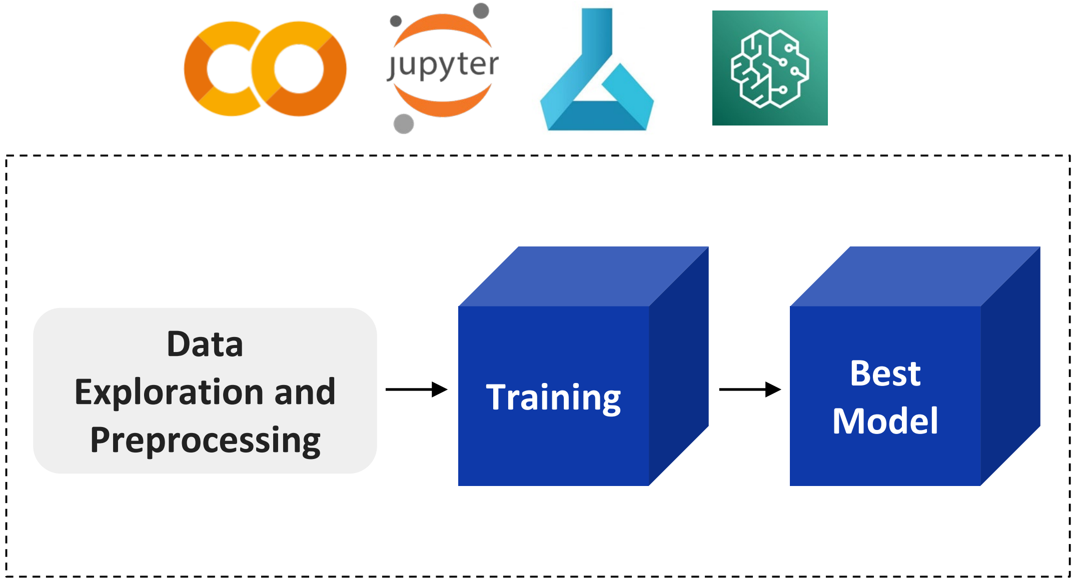

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- What if we **shut down** the development environment?
- Need **to execute all data processing steps** and **build the best model again** from scratch.
- Loss of resources and time.
- We need a way to **"save" the best model** to save our progress.

This is where **serialization** comes in.

Video Break

Model Serialization

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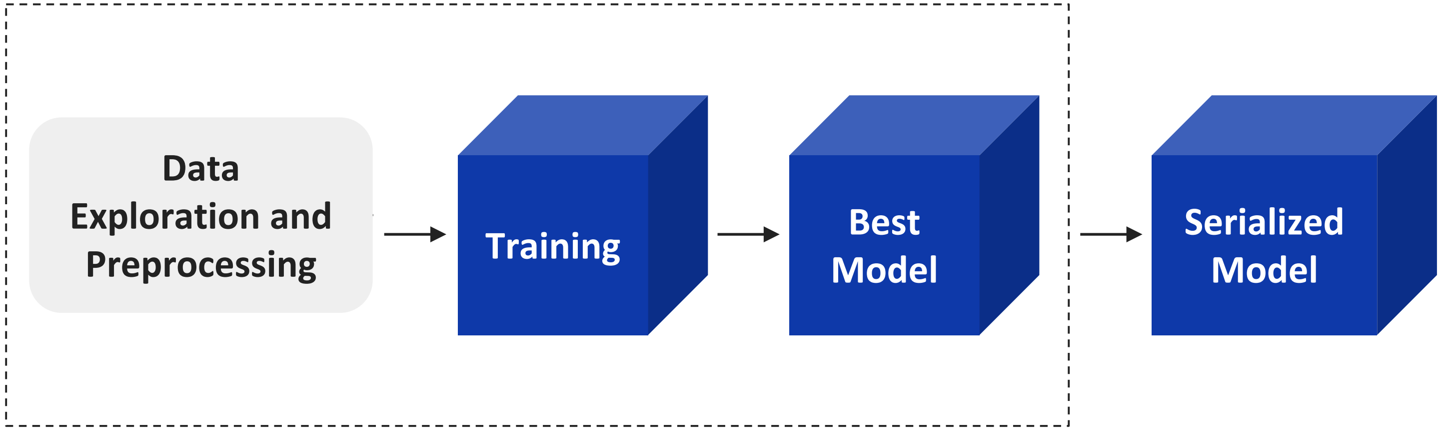

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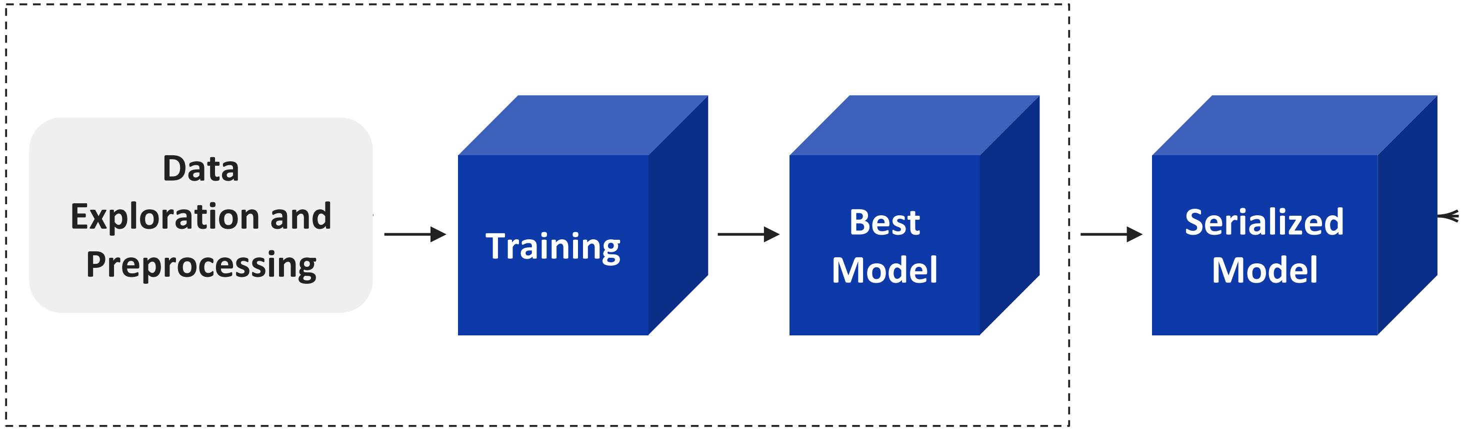

Development Environment

- Can **load** it in **any environment**
- **Need not execute** the entire **code again**
- Can make **faster predictions**

Model Serialization Formats

There are **multiple formats** available for **serializing an ML model** to ensure **compatibility**, **efficiency**, and **ease of use**:

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Format	Use
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Pickle	Saving and loading small models in python for quick and local use
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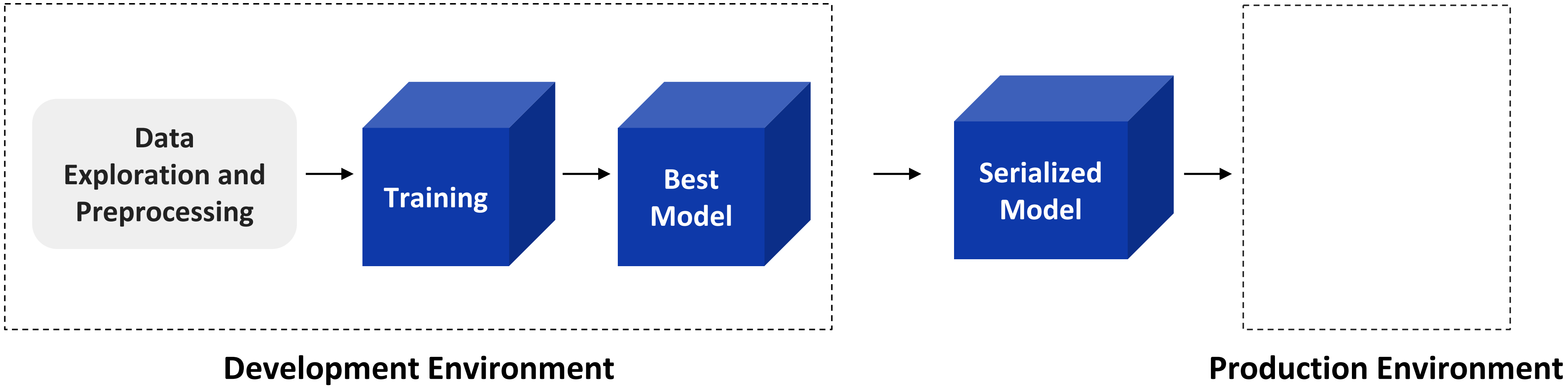
ONNX	Deploying models across different frameworks or running them on edge devices
TensorFlow SavedModel	Deploying deep learning models in production with TensorFlow serving or TensorFlow lite

Serialization



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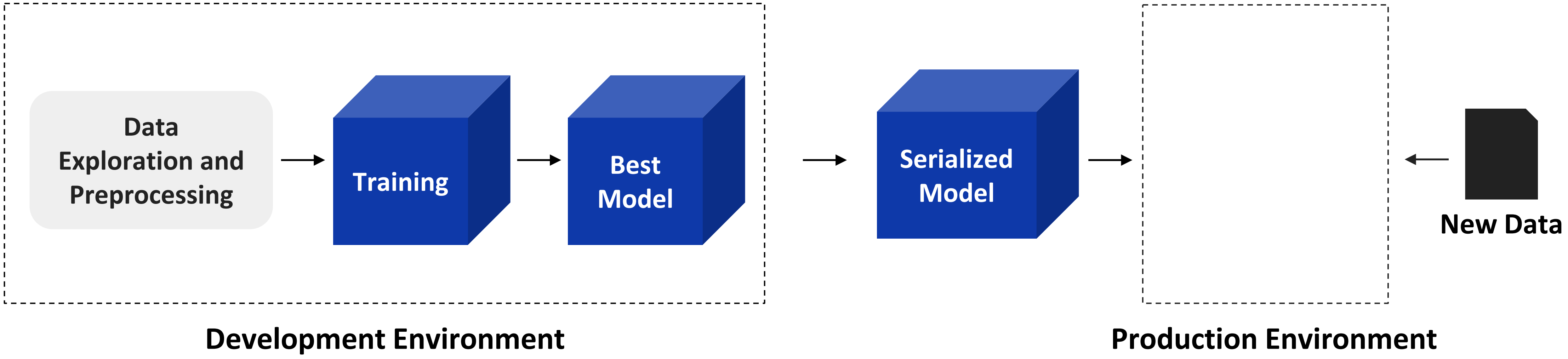

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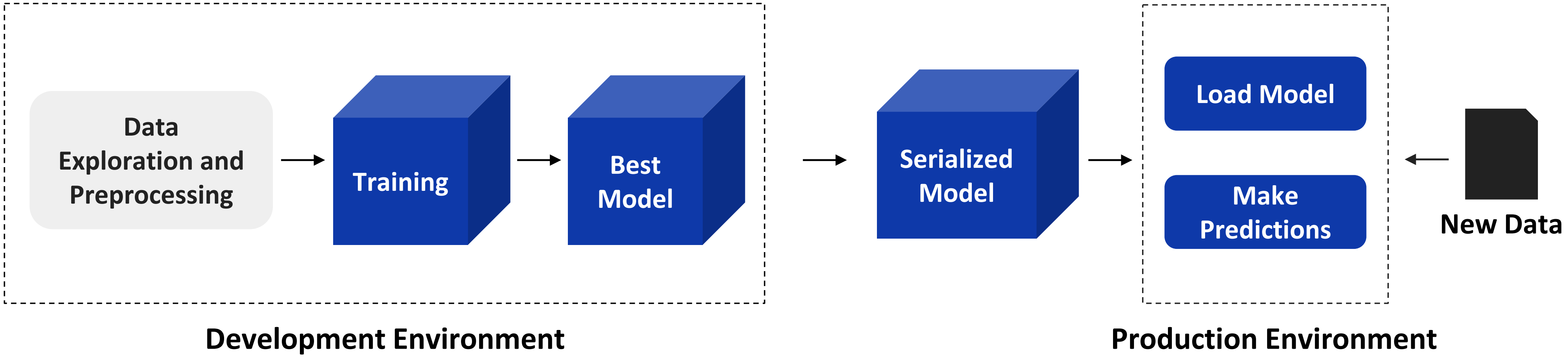

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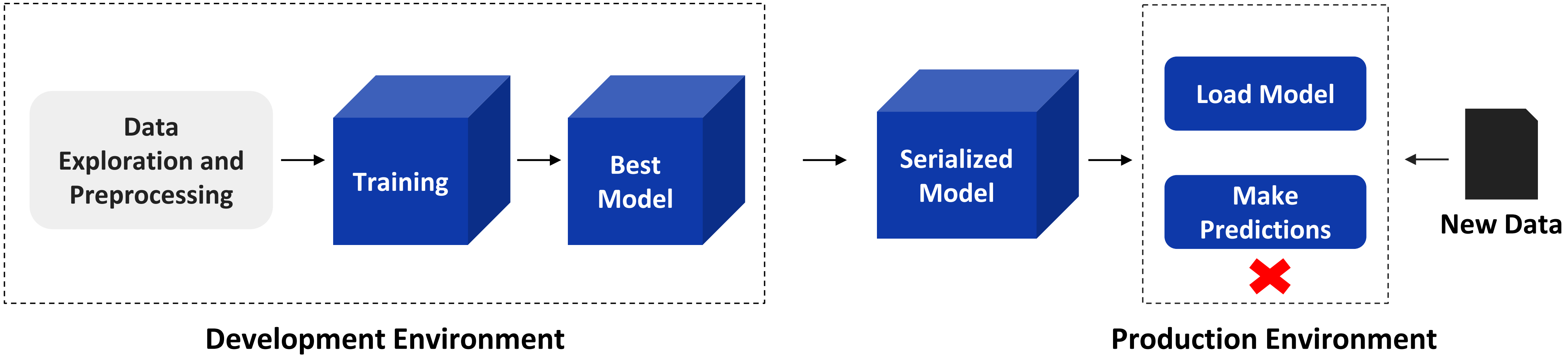
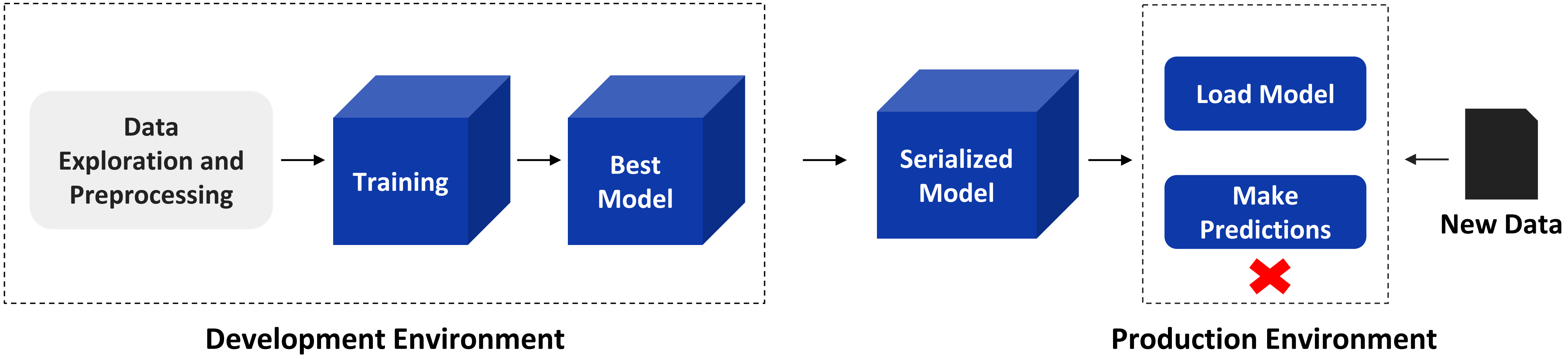


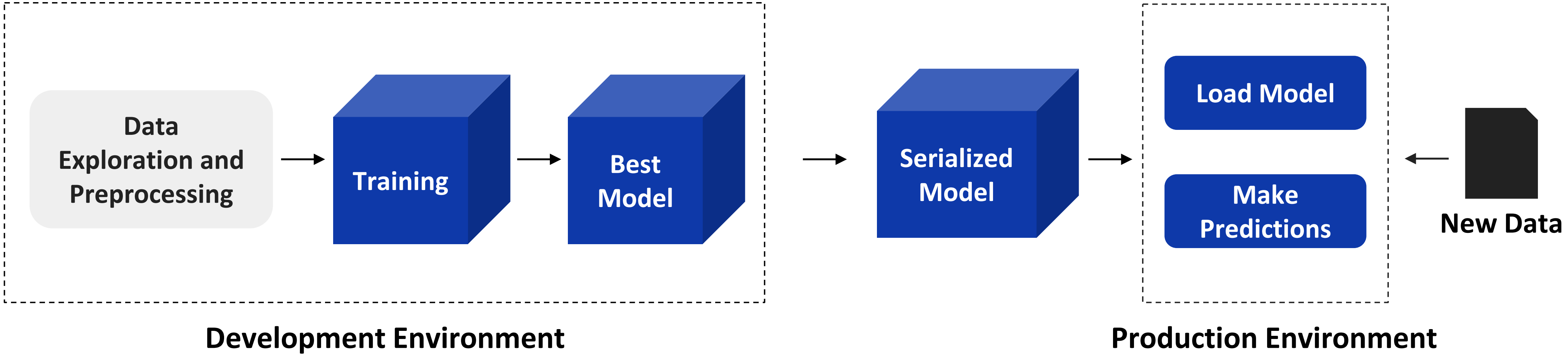
- Why did the predictions fail?

Serialization



- The **new data had additional genres** (say “Thriller” and “Horror”) beyond what was used to train the model (say “Action”, “Adventure”, “Sci-Fi”, “Romance”, and “Comedy”).

Serialization



- We need to **serialize the best model along with the data preprocessing steps** that were used to preprocess the data used to train the model.

Video Break

Addressing Notebook Sharing Challenges

A way to **share** the model

A way to **ensure consistent configurations**

Interface for communication to get fast predictions

Concurrent access for multiple users

1

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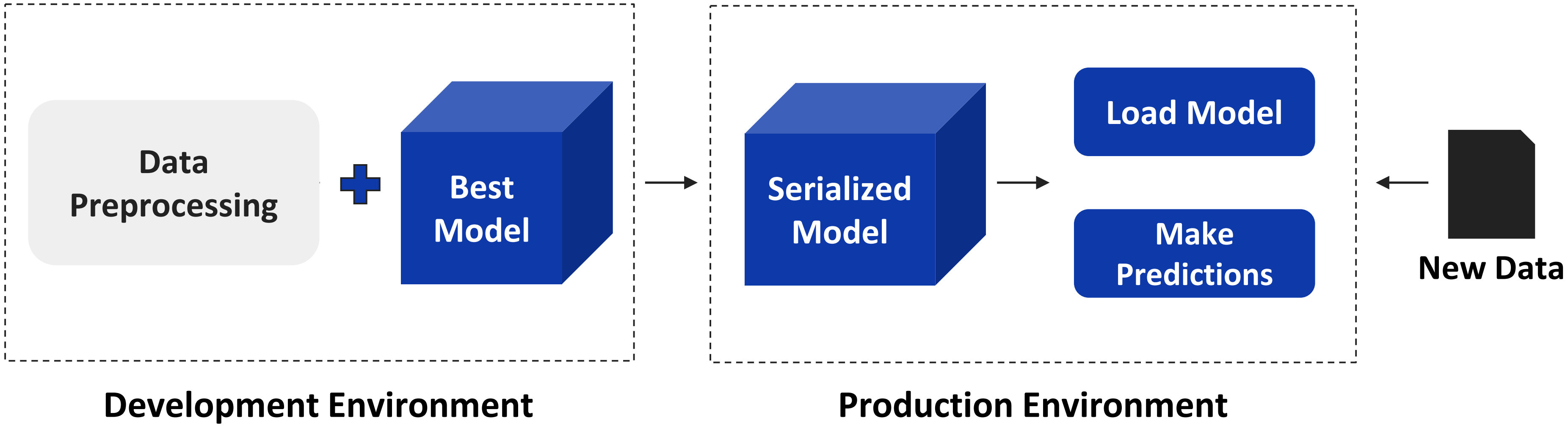
Serialized Model

- Next, let's discuss about the communication interface for faster predictions.

The Need for a Communication Interface



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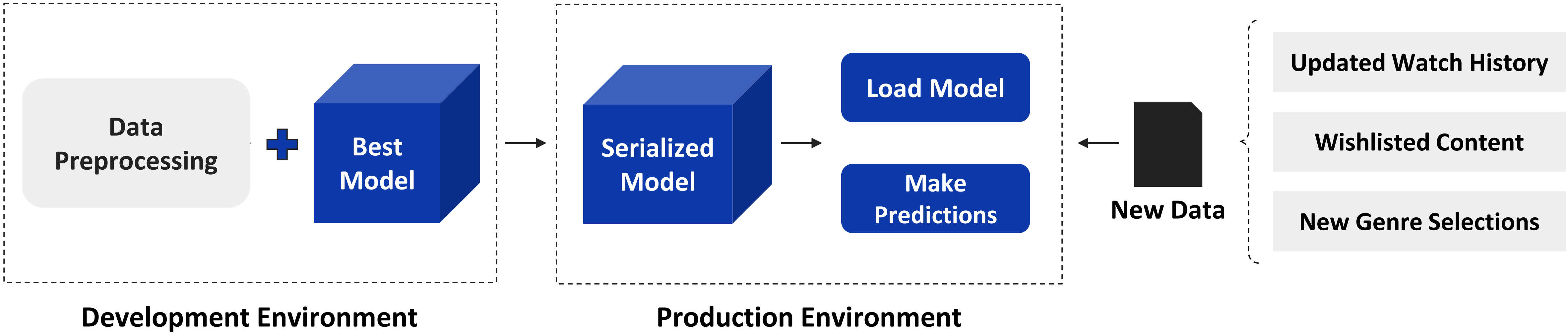

- How can the model receive the new user data and then generate recommendations accordingly?

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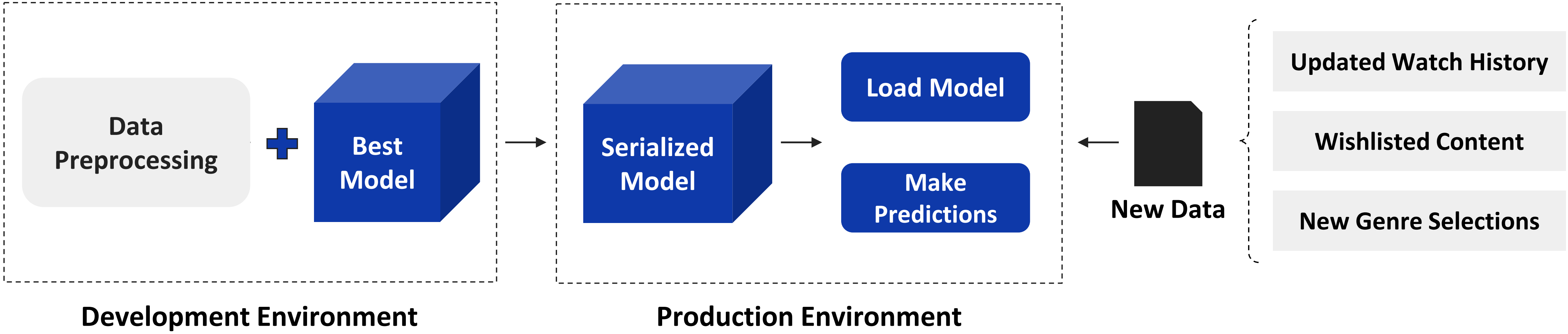
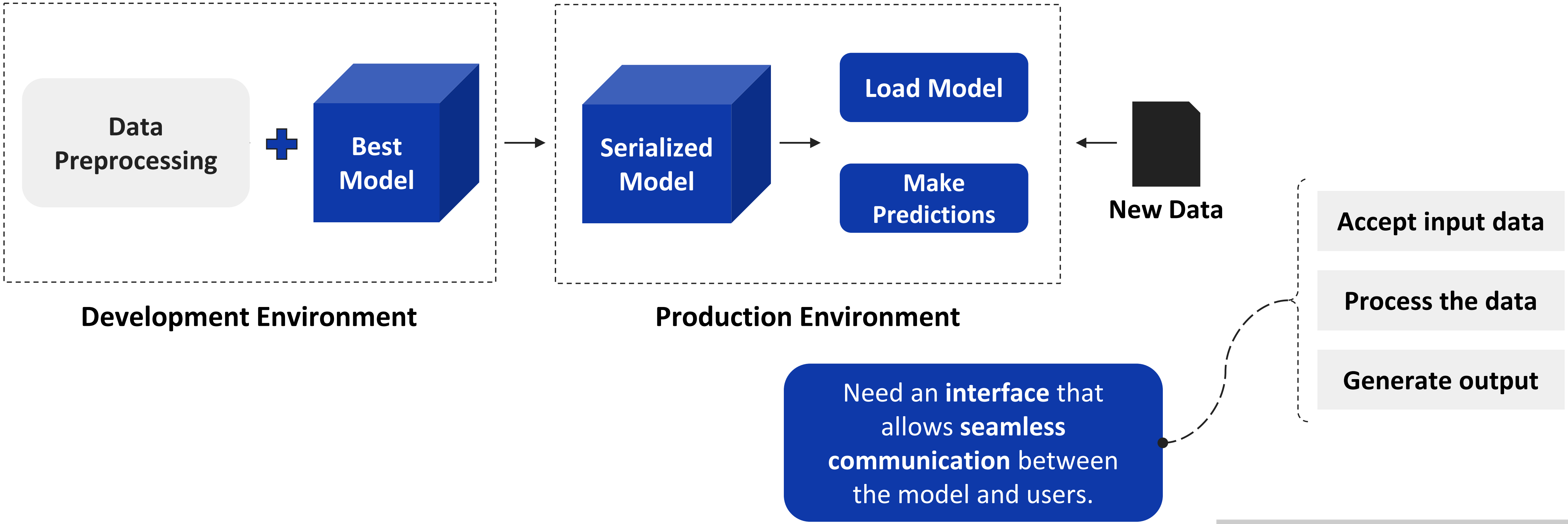


- How can the model receive the new user data and then generate recommendations accordingly?
- Important to note that we **DON'T** want the user **to access the model directly** as the raw output might not be useful to the end user.

The Need for a Communication Interface



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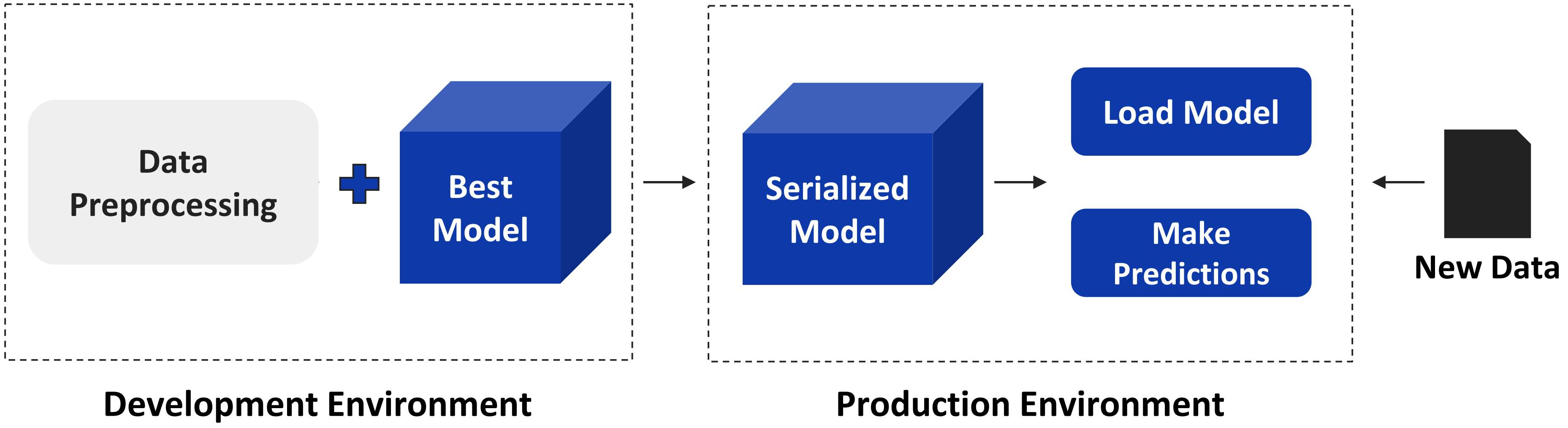

- Accept input data
- Process the data
- Generate output

Need an **interface** that allows **seamless communication** between the model and users.

This is where an **API** comes in.

Application Programming Interface (API)

Application **P**rogramming **I**nterfaces, or **APIs**, are **mechanisms** that **enable two software components to communicate** with each other **using a set of definitions and protocols**.

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Netflix App

Movies

Recommendations

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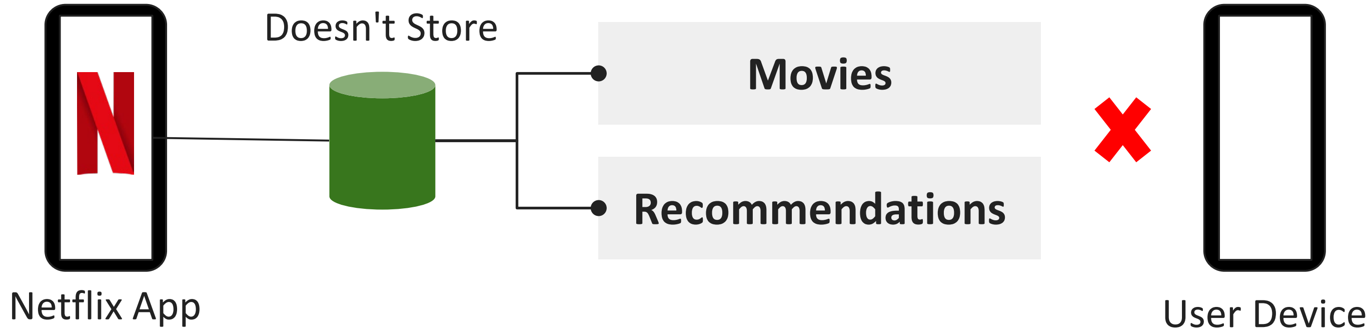

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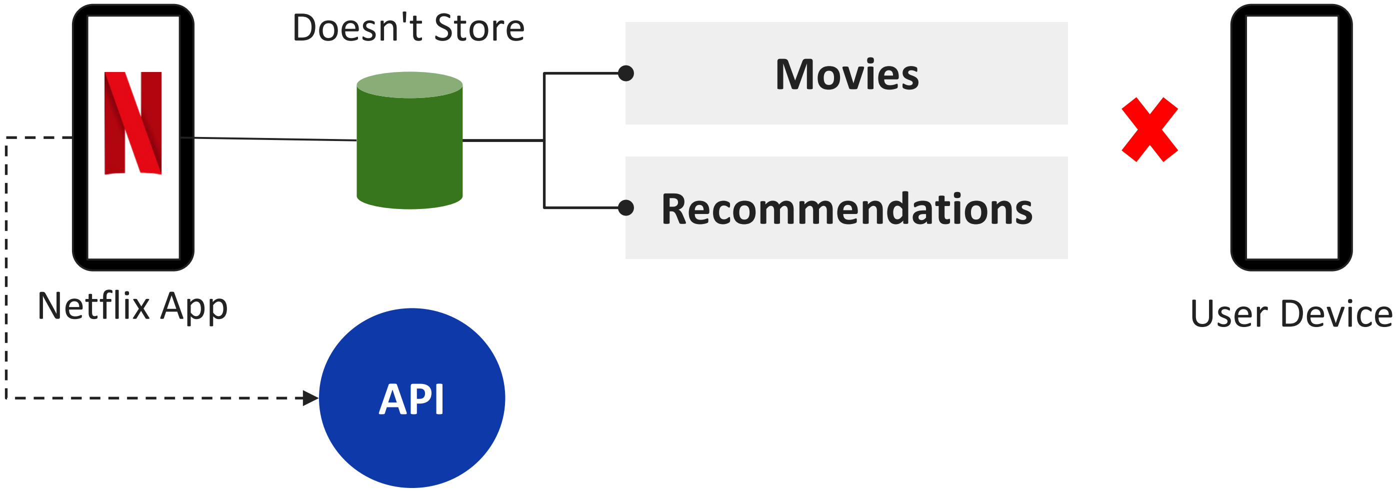

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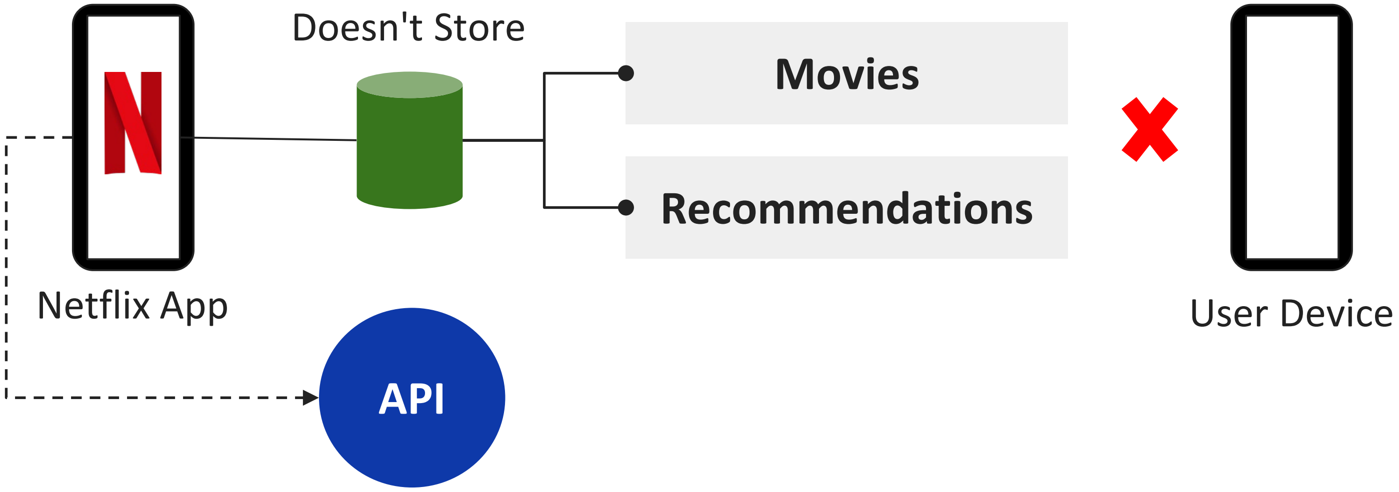
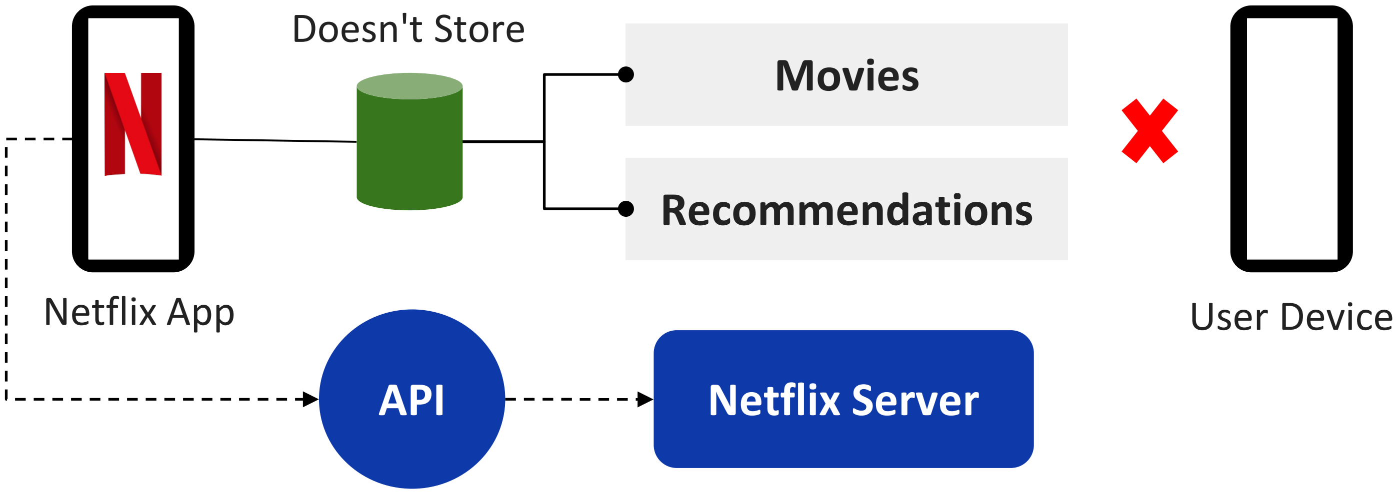


- But then where is the data fetched from?

Server

A **server** is a **central computer** from which other computers get information.



HTTP Methods

- When a user wants to get predictions from a model via an API, they need to send a request.

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Send Data	Profile information from new users.
Update Data	Modifications to user interests or preference.
Delete Data	Clearing user watch history.

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GET	**Retrieve data from an API**	Get a movie recommendation.
POST	**Send data to the API**	Send current watchlist to the model.
PUT	**Update existing data**	Add new movie to the watchlist.
DELETE	**Remove data**	Clear watch history.

- Note that all these operations **DON'T** require an ML model.

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- **HTTP status codes** provide this information, ensuring smooth communication between users and the API.

HTTP Status Codes

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401	Unauthorised	Authentication is required.
403	Forbidden	Access is not allowed.
404	Not Found	Requested resource does not exist.
500	Internal Server Error	The server encountered an error.

Video Break

Addressing Notebook Sharing Challenges

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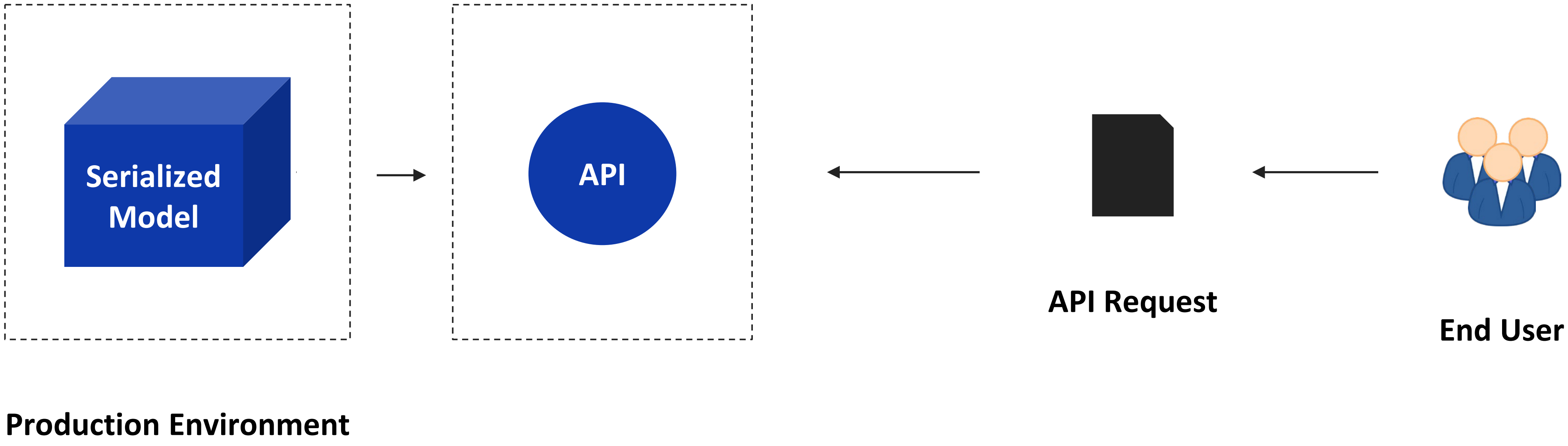
A way to **share** the model	A way to **ensure consistent configurations**	**Interface for communication** to get fast predictions	**Concurrent access** for multiple users
Serialized Model		**API**	

- Let’s now see how can we solve the challenge of “concurrent access for multiple users”.

The Need for Endpoints

- A user made an API request to get a movie recommendation.



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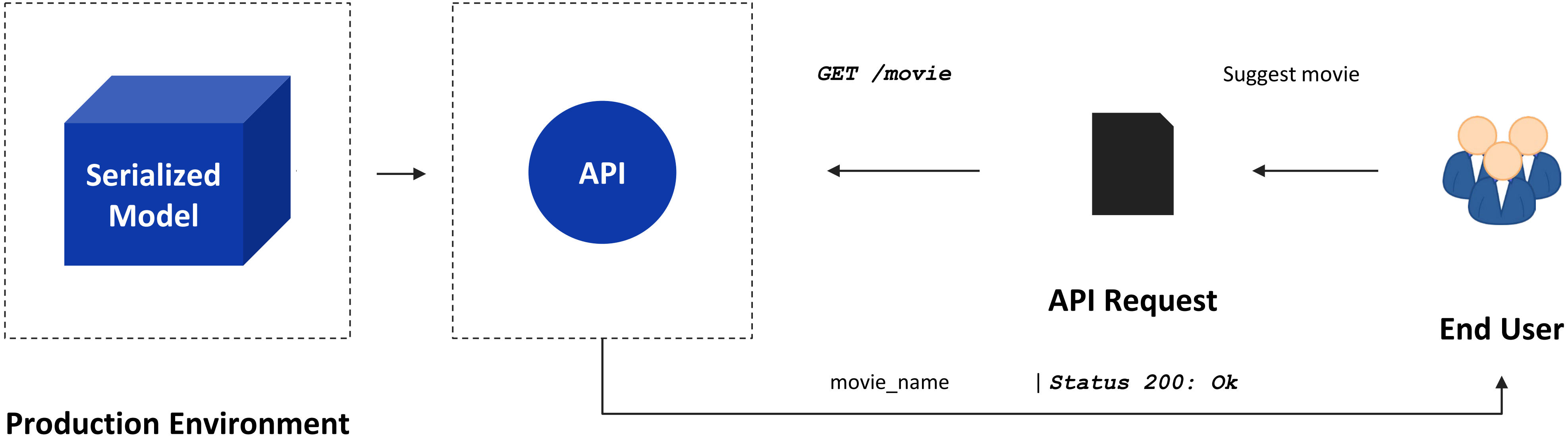
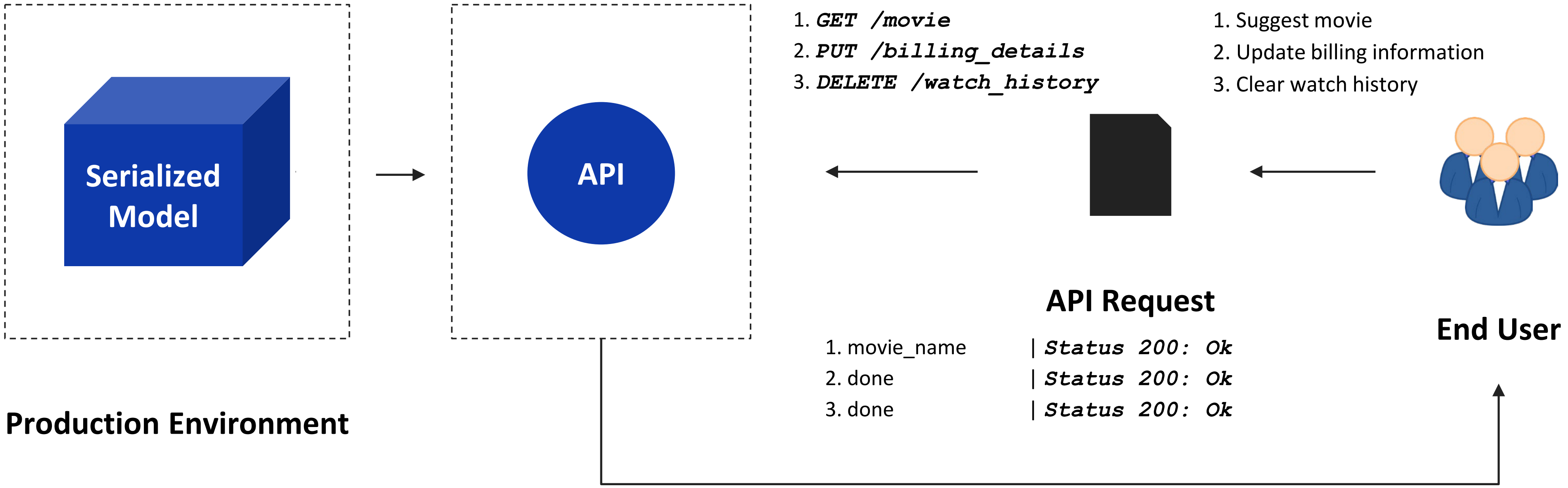


- The **instruction** to the API and the **response** from the API are **technical** in nature.

The Need for Endpoints

- A user can make multiple requests for multiple tasks.



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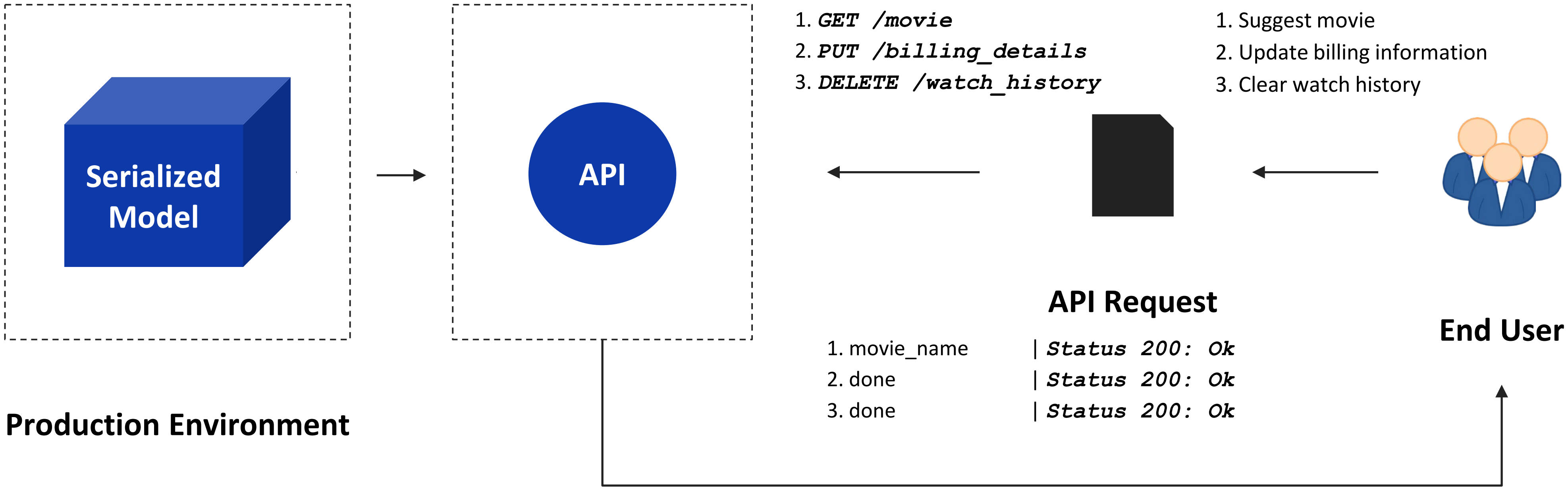
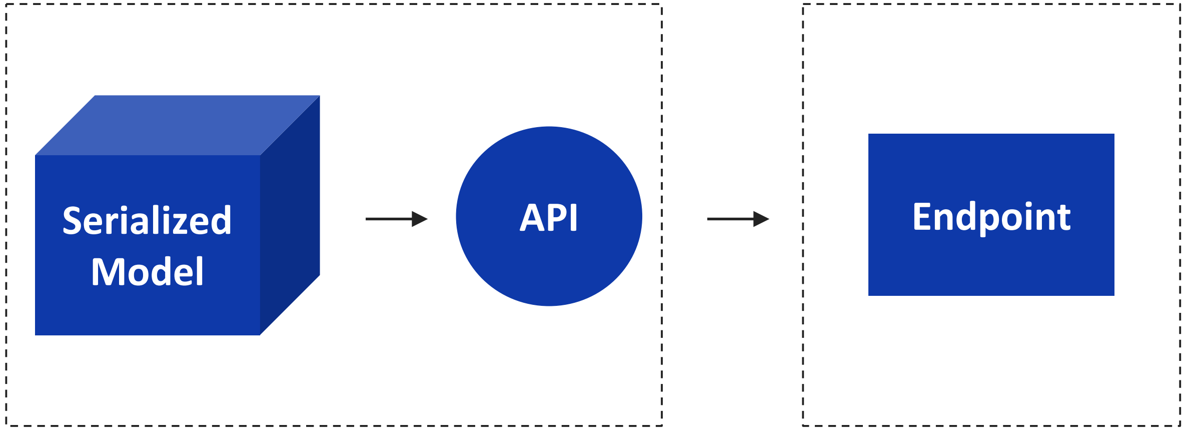


- It’s **not feasible** for a user to **remember** all **the instructions and status codes.**

Endpoints

- We can **create an endpoint from an API** to make things **simpler for the end user**.



Production Environment

Endpoints

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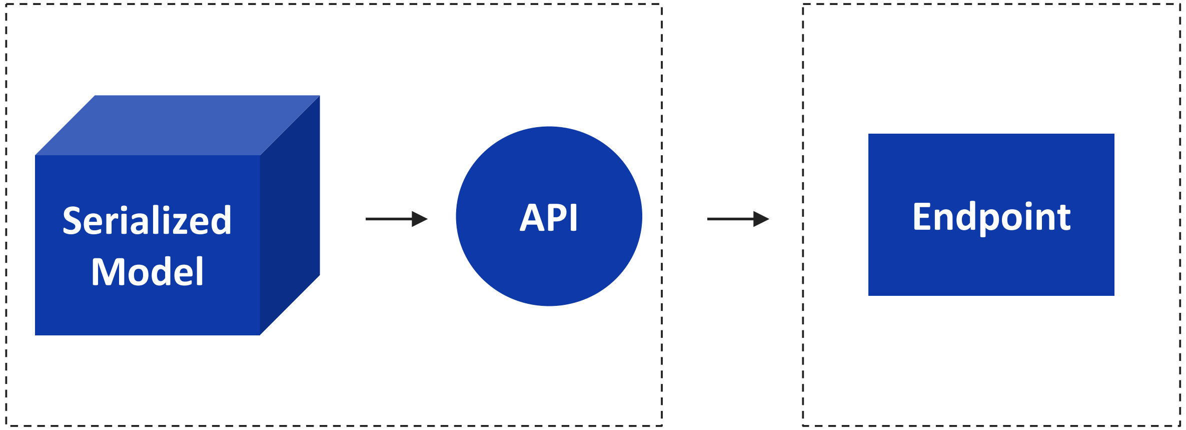
An **API endpoint** is a **digital location** where an **API receives API requests**, for resources on its server.



Production Environment

Endpoints

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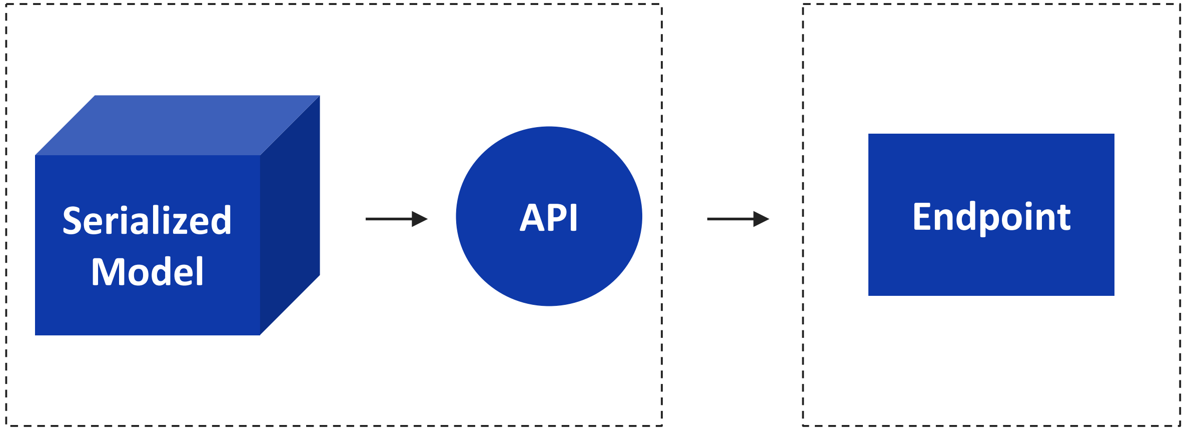
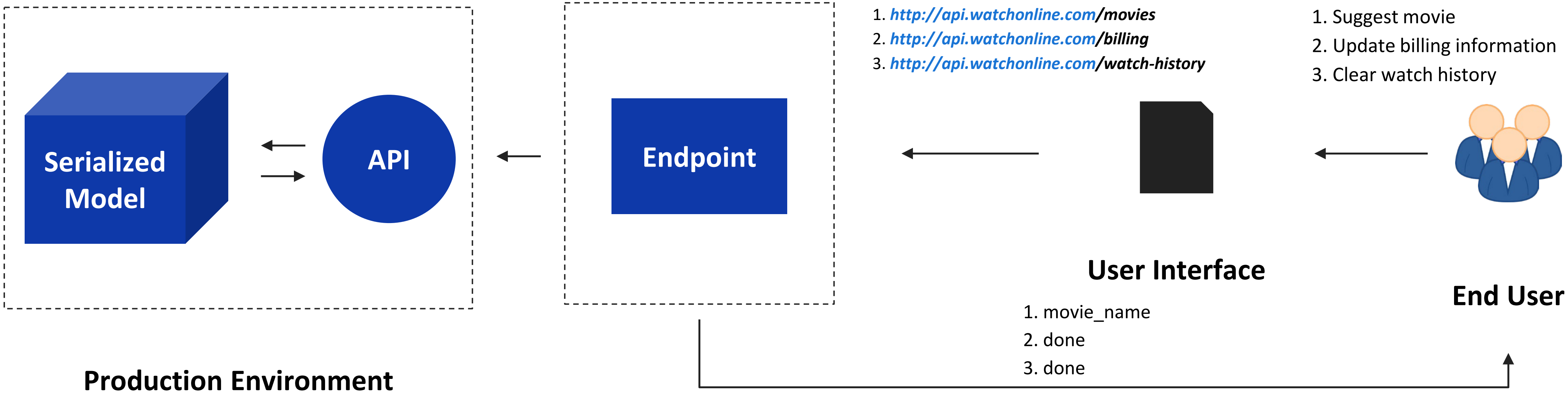
- They are most often in the form of URLs (**U**niform **R**esource **L**ocators).



Production Environment

Endpoints

- Each functionality (API request) is performed via an endpoint (URL), which is generally a user interface.



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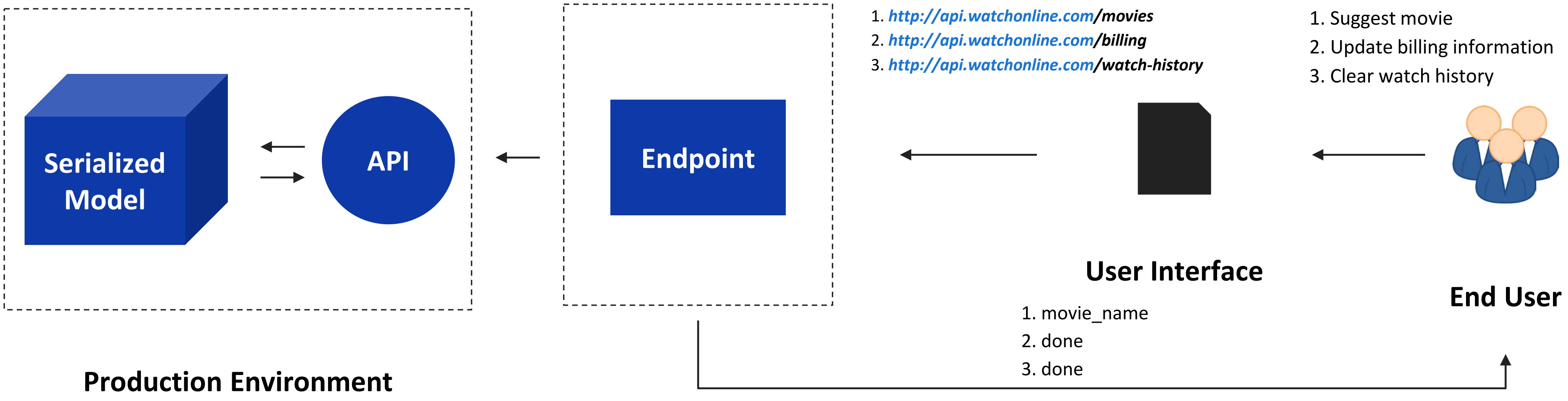

- Note that **a part of the URL remains the same**, while **other parts change as per the user request.**

Hosting an Endpoint

- How do we create an endpoint that allows multiple users to interact with the model?

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Hosting is the **ability** to make applications and websites available on the internet.

- There are multiple ways to host an endpoint, but the two most common ways are the following:

Local Machine	**Self-hosting the model on a local server, accessible only within a restricted environment.**
Cloud Platform	**Hosting the model on cloud platforms (like AWS, Azure, or GCP) making it accessible globally.**

Video Break

Addressing Notebook Sharing Challenges

A way to **share** the model

A way to **ensure consistent configurations**

Interface for communication to get fast predictions

Concurrent access for multiple users

1 **Serialized Model**

2

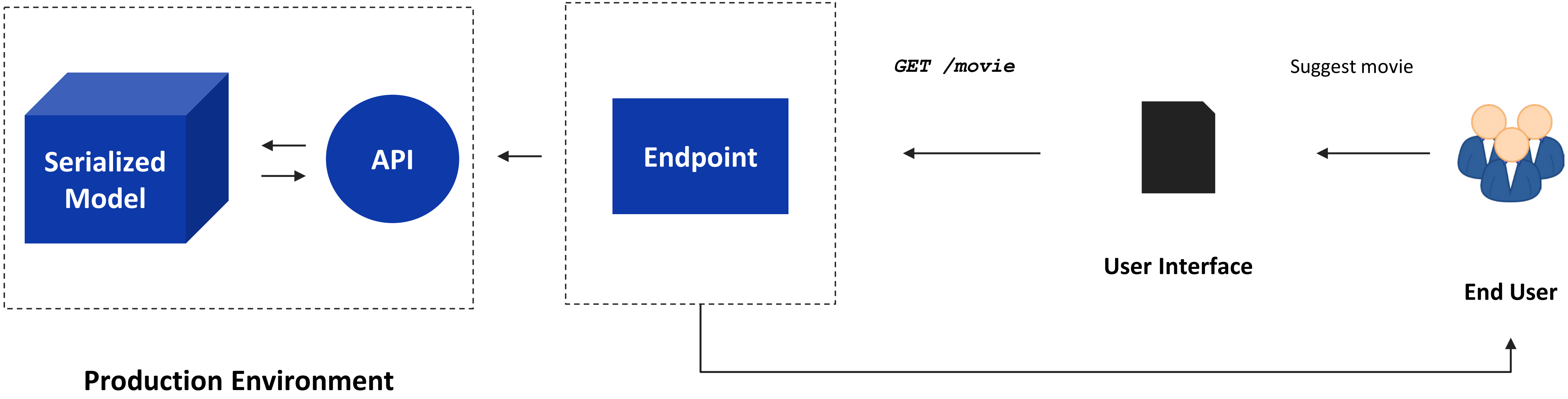
3 **API**

4 **Hosted Endpoint**

- Now, let's solve the challenge of consistent configurations (environments).

Role of Dependencies

- A user tried to get a movie recommendation from the API endpoint.



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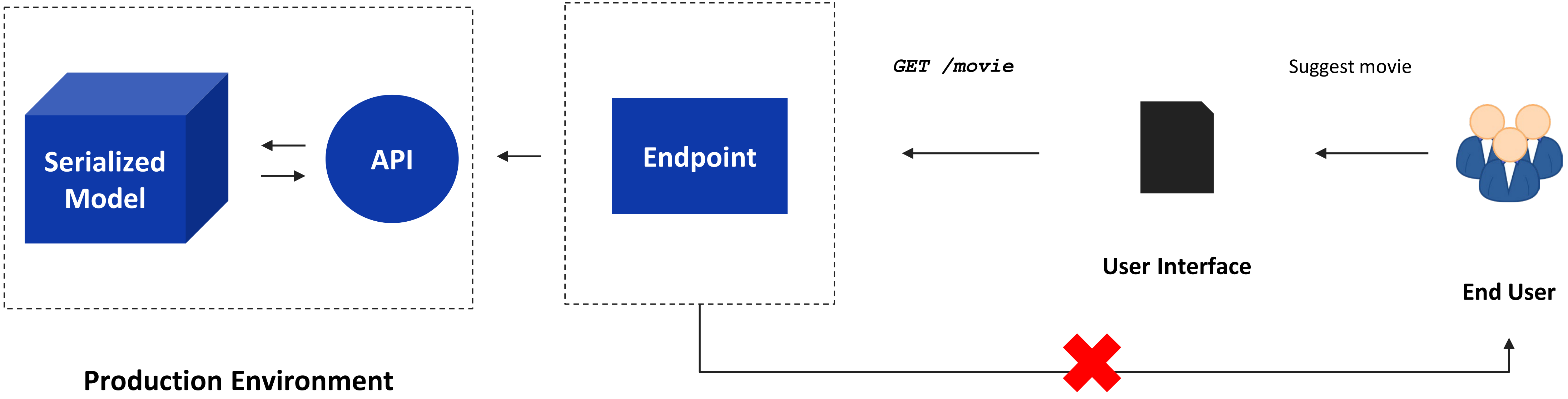

- Let's see why the user failed to get a response.

Role of Dependencies

Has the serialized model and the API

Production Environment

✘

Deployment failed

▼

Role of Dependencies

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Developer tried to install the missing library in the production environment

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- This is inefficient way to handle dependencies every time there's a failure.

Handling Dependencies

- What if we keep all dependencies at a single place and take care of them at one go?

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- In general, dependencies files are specified as a text file **(requirements.txt)**.

requirements.txt

```
sklearn==1.6.1
pandas==2.2.3
numpy==2.2.4
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..
```

Handling Dependencies

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- In general, dependencies files are specified as a text file **(requirements.txt)**.
- This creates a stable and reproducible environment, allowing the model to be deployed seamlessly and function as expected in production environment.

requirements.txt

```
sklearn==1.6.1
pandas==2.2.3
numpy==2.2.4
..
..
```

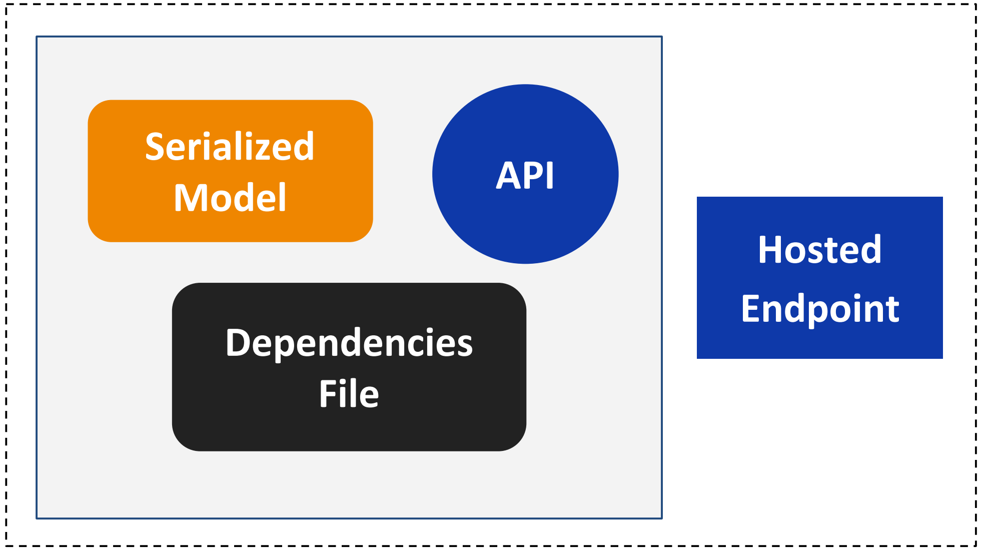
Video Break

Addressing Notebook Sharing Challenges

A way to **share** the model	A way to **ensure consistent configurations**	**Interface for communication** to get fast predictions	**Concurrent access** for multiple users
1	2	3	4
Serialized Model	**Dependencies File**	**API**	**Hosted Endpoint**

- Now that we have resolved all the challenges, we can deploy the model to make it accessible for users who want to make inferences.

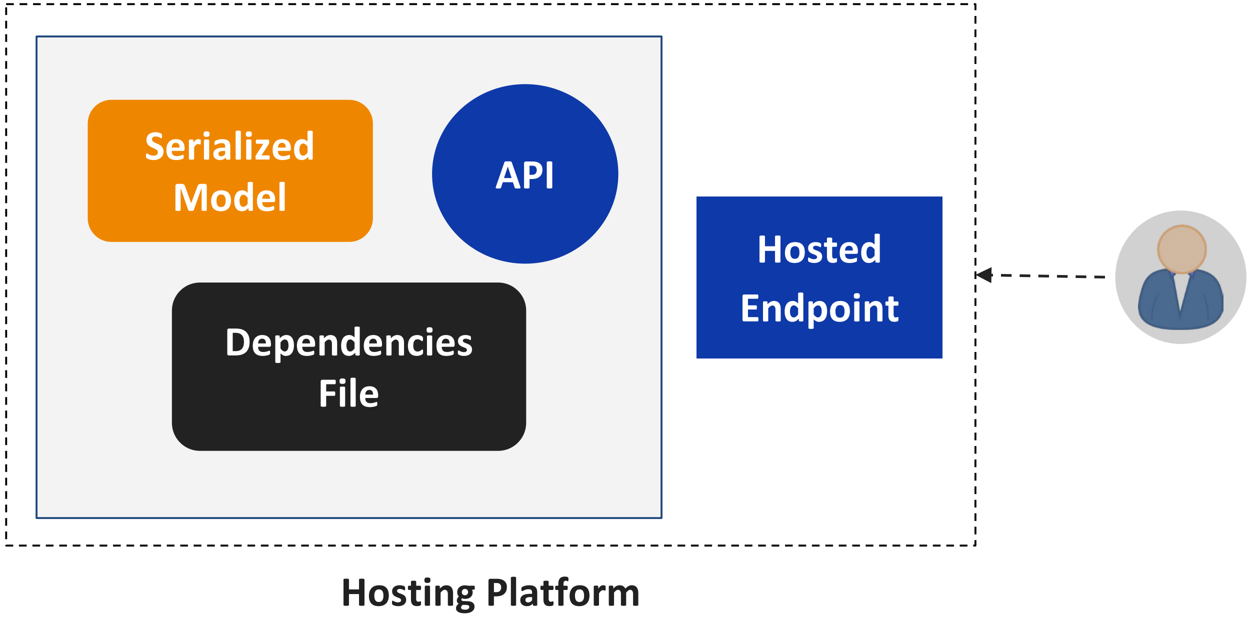
Model Deployment



Hosting Platform

The Need for Security

- Consider the case where a user creates multiple fake accounts and starts sending too many requests.



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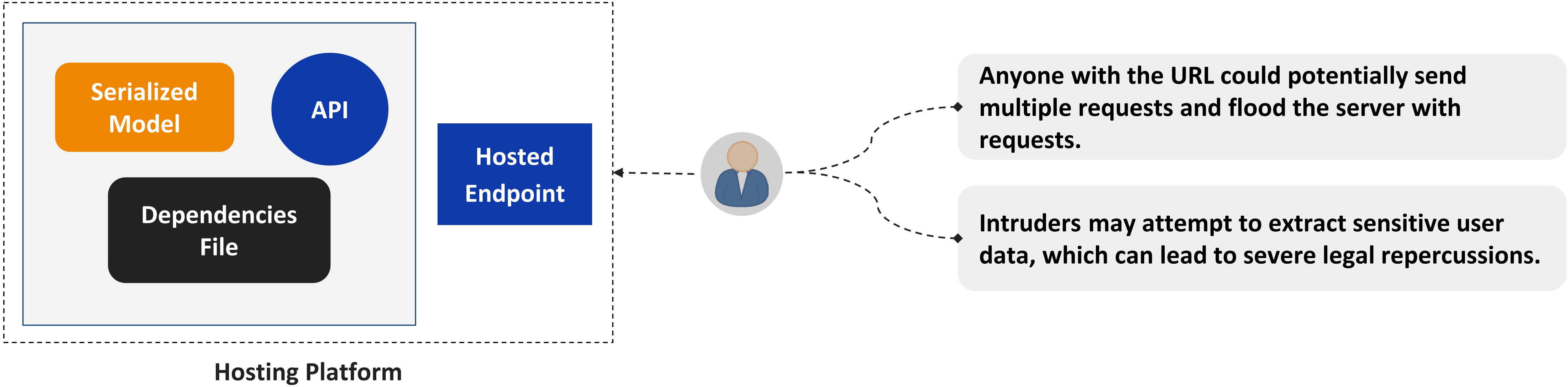

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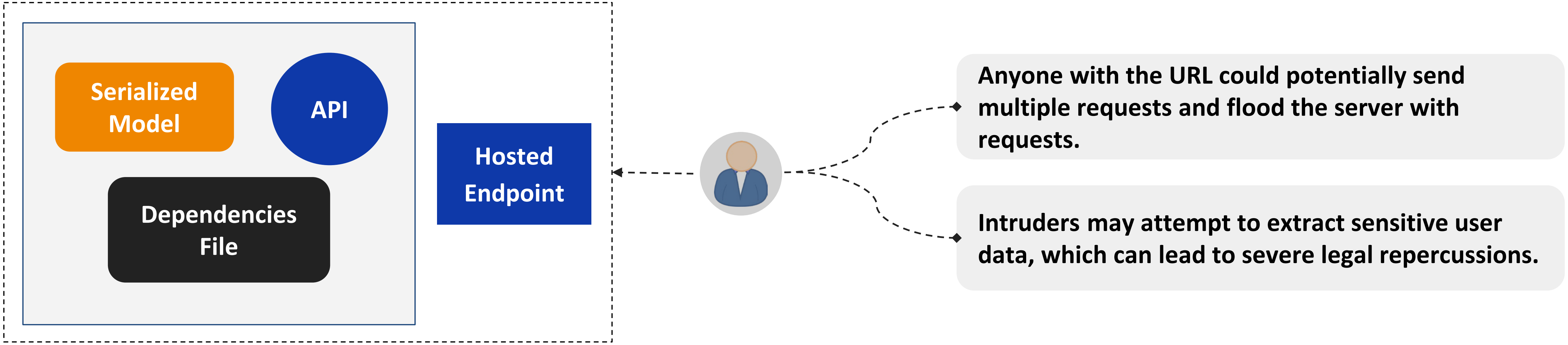

Hosting Platform

- How do we **ensure only trusted users interact with the service** (recommendation model)?

Access Keys

- This is where **access keys** come in.

An **access key** is a **unique identifier used to authenticate and authorize** a user or application to **access a secure resource.**

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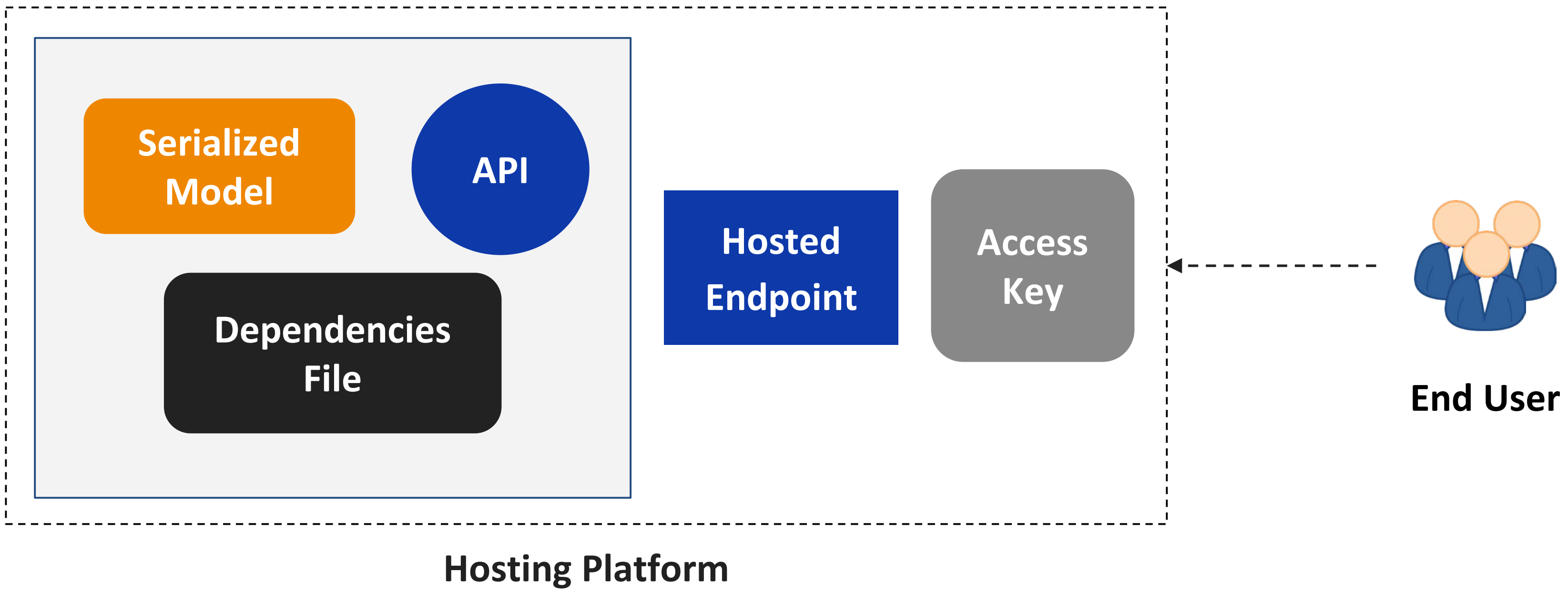
An **access key** is a **unique identifier used to authenticate and authorize** a user or application to **access a secure resource.**

- **Access keys prevent unauthorized** access to secure resources (like data, models, and APIs).
- Only authorized users can use the resources and functionality associated with a specific account.

Secure Model Deployment

- Only authorized users will be able to securely access the public URL (endpoint).



Video Break

Architecture of Model Deployment

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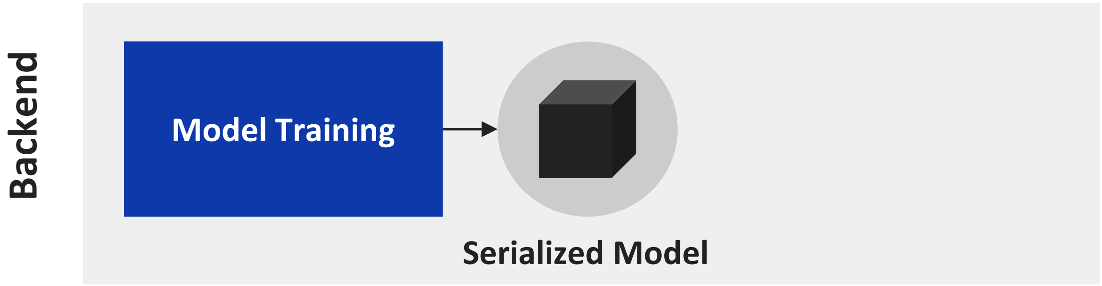

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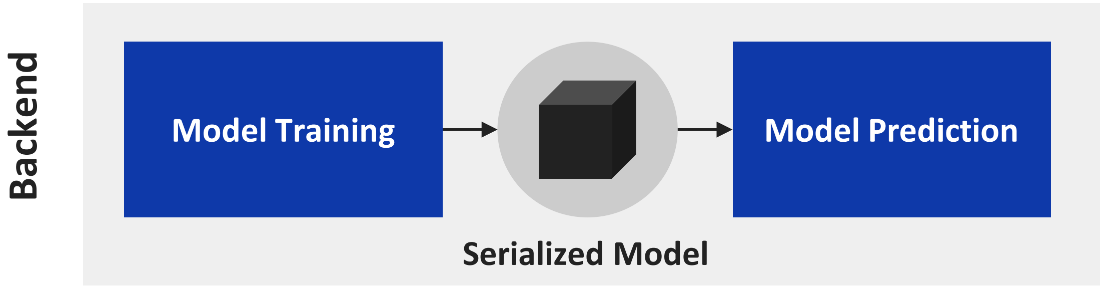
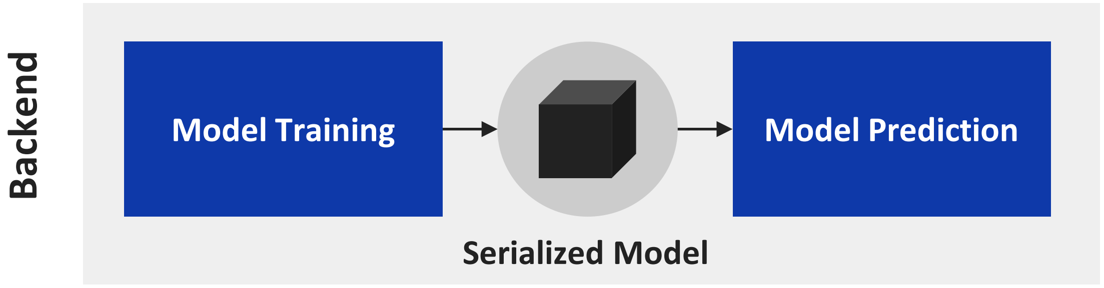


Architecture of Model Deployment

HTTP API Request



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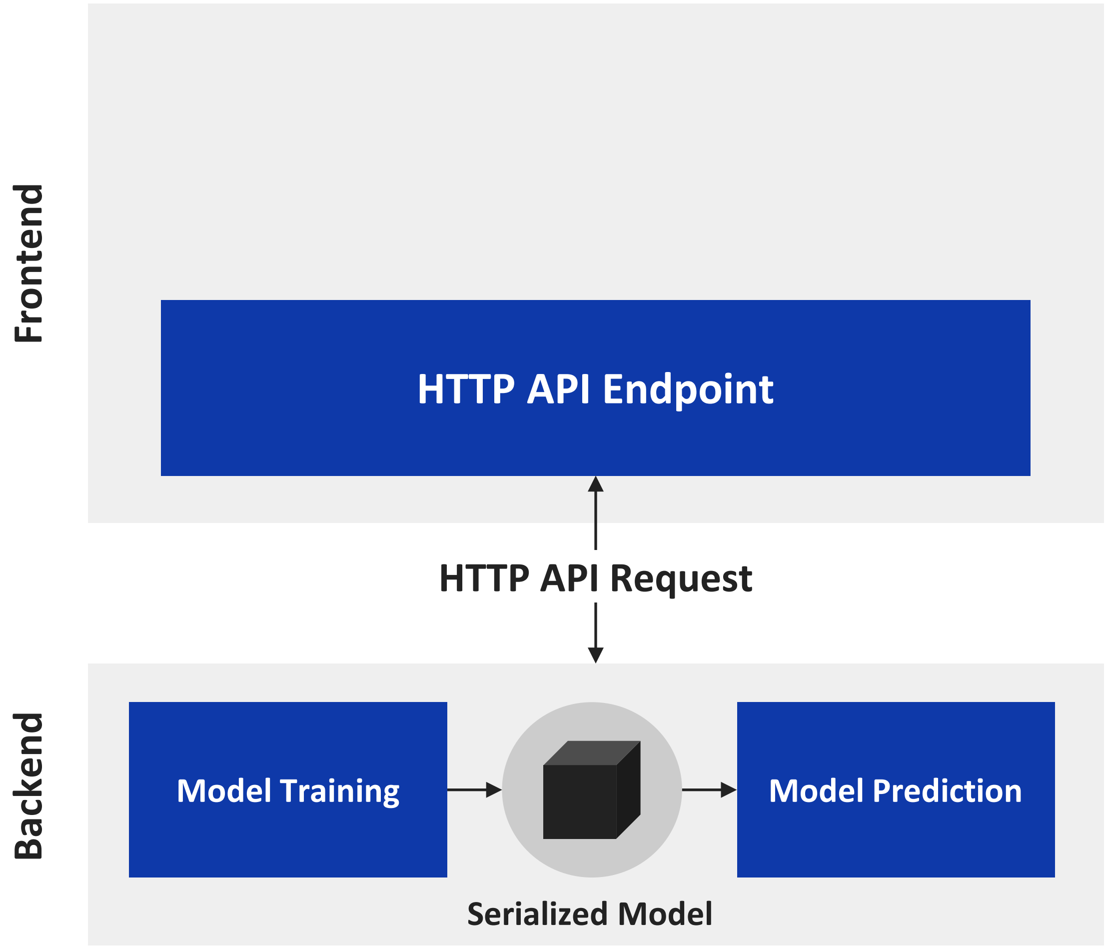

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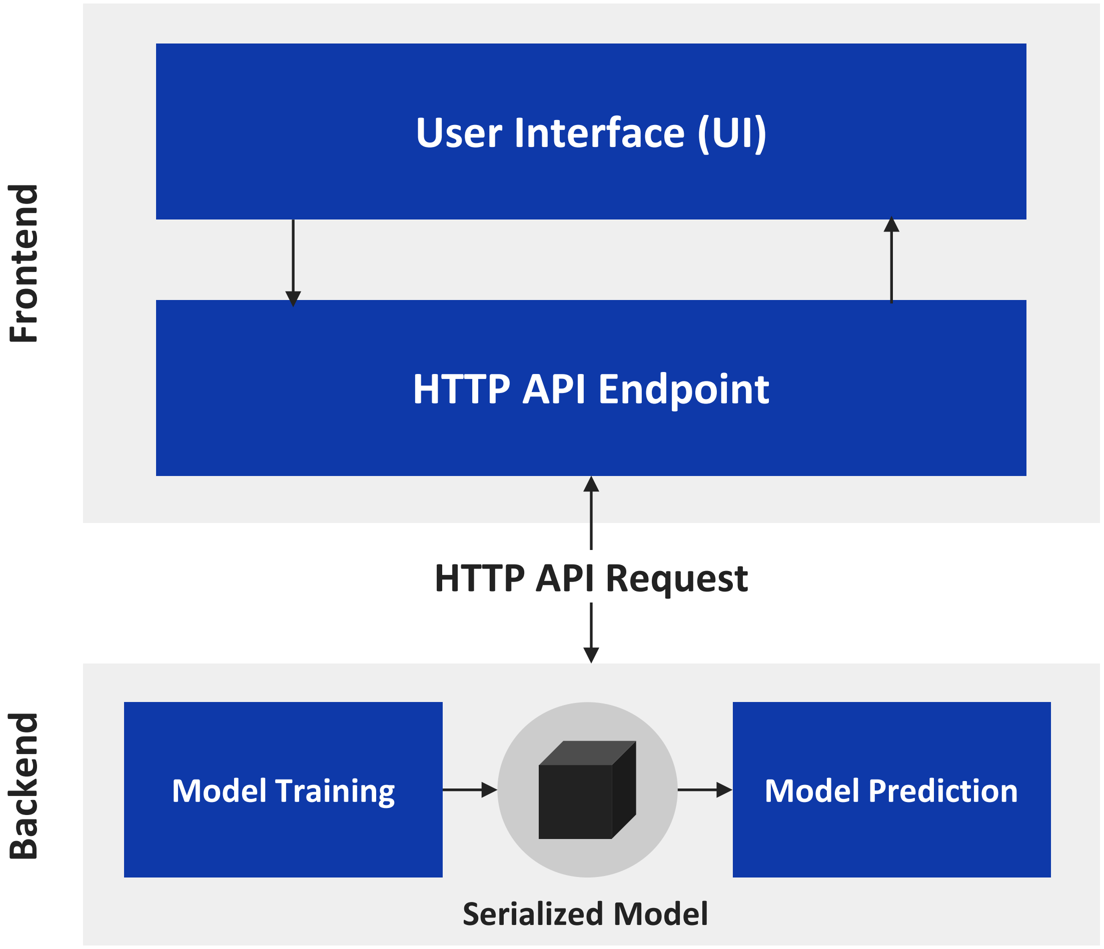

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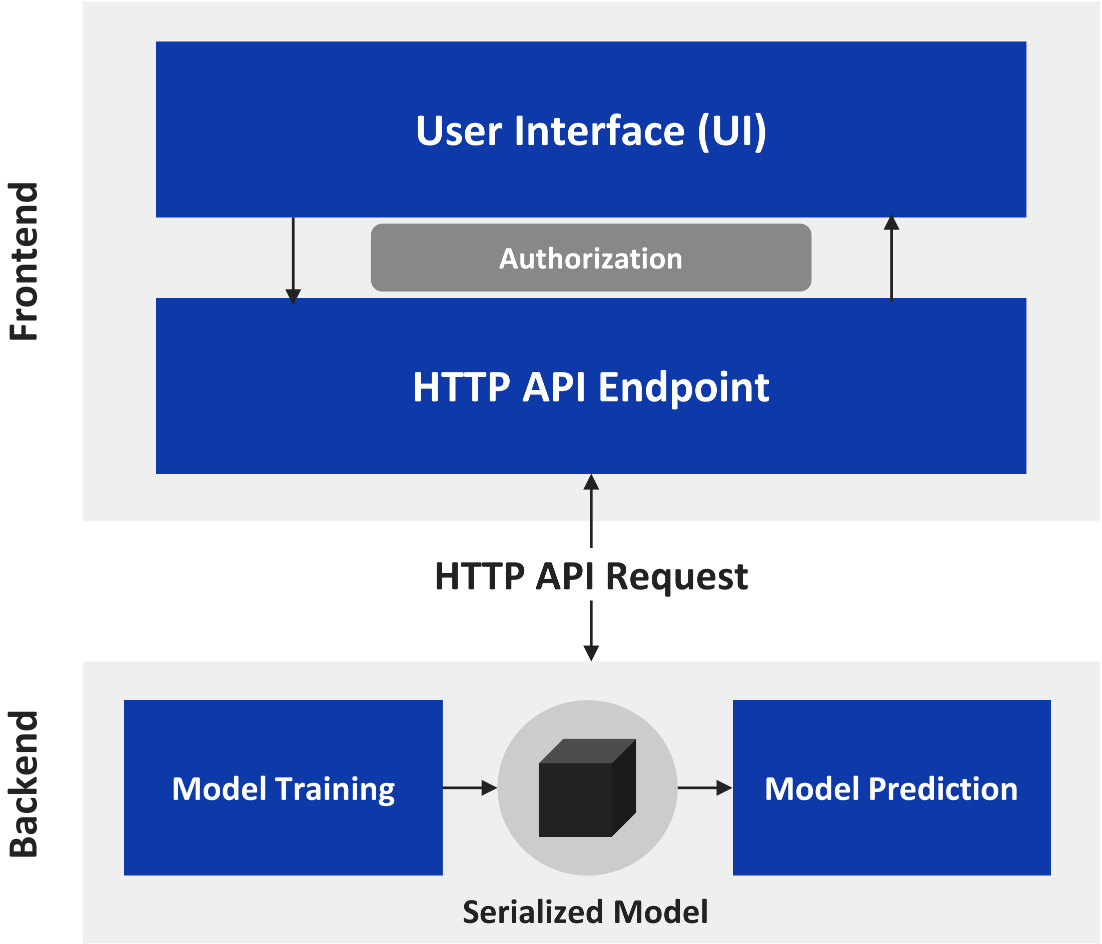

Video Break

Quick Recap

- **Problems Solved using ML**
- **Model Building Process**
- **Need for Deployment**

Scope of ML

Quick Recap

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The Challenges
- Sharing and Consistency
- Speed and Scalability

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- **Architecture of Deployment**
  - Backend
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Learning Outcomes

You should now be able to:

- Summarize the need for model deployment and its role in operationalizing machine learning solutions
- Implement serialization techniques to efficiently store and transfer trained models
- Develop API endpoints to facilitate seamless interaction with deployed models
- Manage dependencies and hosting environments to ensure reliable and scalable model deployment

Power Ahead!